RAJALAKSHMI ENGINEERING COLLEGE

Hema Varshini K 251001076-2025-IT

H2

Started on	Friday, 31 July 2026, 3:36 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:30 PM
Time taken	26 days 7 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void function(int n) {
    int count = 0;

    int i = 1; count++;
    int s = 1; count++;

    while(s <= n) {
        count++;

        i++;  count++;
        s += i; count++;
    }
    count++;

    printf("%d\n", count);
}

int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        function(n);
    }
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Hema Varshini K 251001076-2025-IT

H2

Started on	Tuesday, 22 September 2026, 9:09 AM
State	Finished
Completed on	Tuesday, 22 September 2026, 9:13 AM
Time taken	4 mins 38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void func(int n)
{
    int count = 0;

    if(n == 1)
    {
        count = 1;
    }
    else
    {
        for(int i = 1; i <= n; i++)
        {
            count += 5;

            for(int j = 1; j <= n; j++)
            {
                break;
            }
        }

        count += 2;
    }

    printf("%d", count);
}

int main()
{
    int n;
    scanf("%d", &n);
    func(n);

    return 0;
```

```
36 }
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Hema Varshini K 251001076-2025-IT

H2

Started on	Tuesday, 22 September 2026, 9:14 AM
State	Finished
Completed on	Tuesday, 22 September 2026, 9:23 AM
Time taken	9 mins 20 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int num;
    scanf("%d", &num);

    int count = 0;

    count++;  // i = 1

    for (int i = 1; i <= num; ++i) {
        count++;  // if condition

        if (num % i == 0) {
            count++;  // printf("%d ", i)
        }

        count++;  // ++i
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Hema Varshini K 251001076-2025-IT

H2

Started on	Tuesday, 22 September 2026, 9:20 AM
State	Finished
Completed on	Tuesday, 22 September 2026, 9:20 AM
Time taken	27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    int c = 0;

    scanf("%d", &n);

    int i = n / 2;
    c++;  // outer loop initialization

    while (1)
    {
        c++;  // outer loop condition
        if (i >= n)
            break;

        int j = 1;

        while (1)
        {
            c++;  // middle loop condition
            if (j >= n)
                break;

            int k = 1;

            while (1)
            {
                c++;  // inner loop condition
                if (k >= n)
                    break;

                c++;  // original counter increment

                k = k * 2;
            }

            j = j * 2;
        }

        i++;
    }
```

```
46      printf("%d", c);
47
48      return 0;
49  }
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Hema Varshini K 251001076-2025-IT

H2

Started on	Tuesday, 22 September 2026, 9:22 AM
State	Finished
Completed on	Tuesday, 22 September 2026, 9:22 AM
Time taken	39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    int rev = 0, remainder;
    int c = 0;

    scanf("%d", &n);

    while (n != 0)
    {
        remainder = n % 10;
        c++;

        rev = rev * 10 + remainder;
        c++;

        n /= 10;
        c++;

        c++;
    }

    c += 3;

    printf("%d", c);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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